Customer Rating of Breakfast Cereals

The dataset Cereals.xls includes nutritional information, store display, and consumer ratings for 77 breakfast cereals, with 16 attributes

1. Name: Name of cereal
2. mfr: Manufacturer of cereal where A = American Home Food Products; G = General Mills; K = Kelloggs; N = Nabisco; P = Post; Q = Quaker Oats; R = Ralston Purina
3. type: cold or hot
4. calories: calories per serving
5. protein: grams of protein
6. fat: grams of fat
7. sodium: milligrams of sodium
8. fiber: grams of dietary fiber
9. carbo: grams of complex carbohydrates
10. sugars: grams of sugars
11. potass: milligrams of potassium
12. vitamins: vitamins and minerals - 0, 25, or 100, indicating the typical percentage of FDA recommended
13. shelf: display shelf (1, 2, or 3, counting from the floor)
14. weight: weight in ounces of one serving
15. cups: number of cups in one serving
16. rating: a rating of the cereals calculated by Consumer Reports

Data-preprocessing: Remove all cereals with missing values, do normalization and appropriate attribute selection

Data Mining Steps:

a) Apply K-Mean algorithm with best K
b) Explain the meaning of each cluster by going their description.
c) The elementary public schools would like to choose a set of cereals to include in their daily cafeterias. Every day a different cereal is offered, but all cereals should support a healthy diet. For this goal you are requested to find a cluster of "healthy cereals"

⇨ Submit student_id_creal.pdf & student_id_creal.xml (or ipynb)